Network Operations & Subscriber Risk Analytics

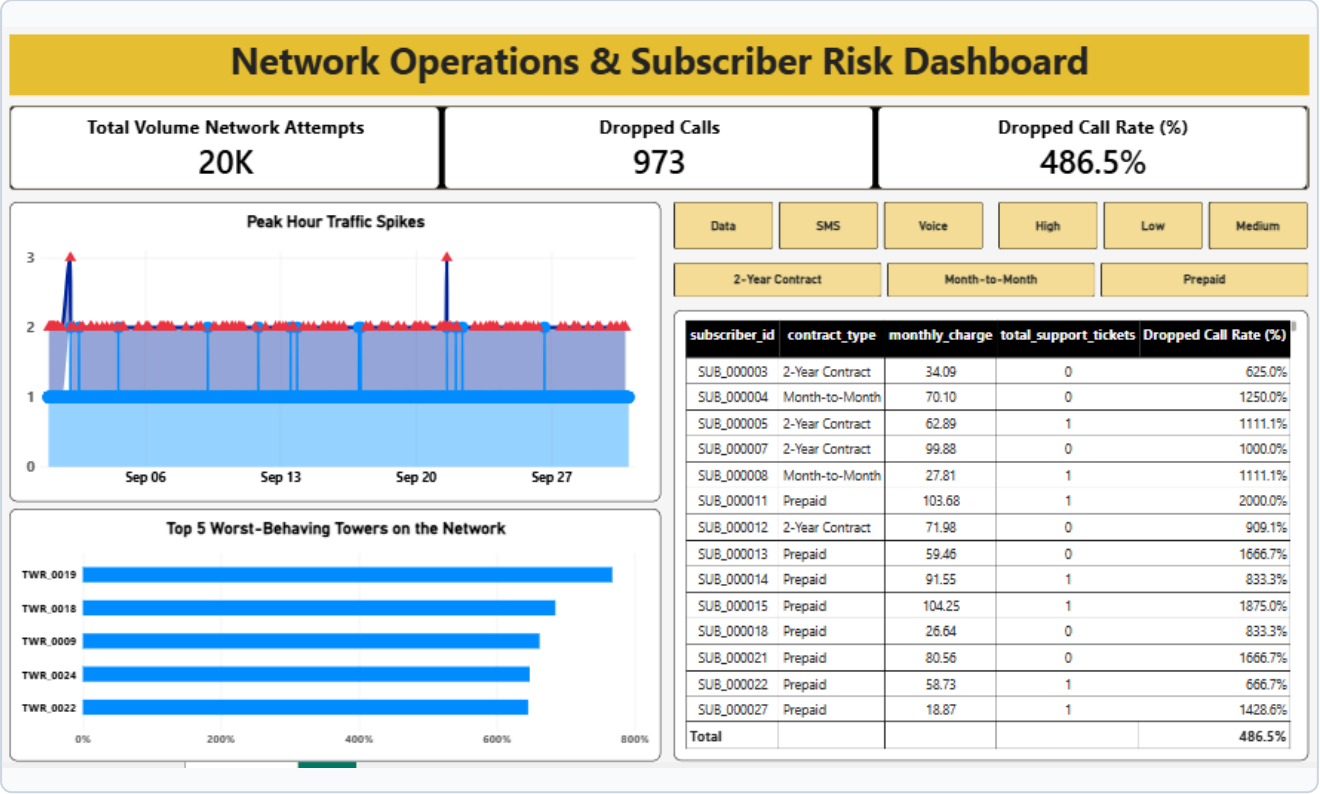
Telecommunications Quality Diagnostics & SQL Data Engineering Report

Author: Data Analytics Team **Domain:** Telecommunications **Stack:** SQL (MySQL/PostgreSQL), Power BI, Python

20K	973	486.5%	TWR_0019
TOTAL ATTEMPTS	DROPPED CALLS	DROPPED CALL RATE	TOP FAILING TOWER

1. Executive Summary & Operations Dashboard

In telecommunications infrastructure management, real-time quality control is crucial for subscriber retention. This project combines operational BI visualization with advanced relational database queries to detect traffic congestion windows, isolate deteriorating cell towers, and identify high-value subscribers facing severe call degradation.



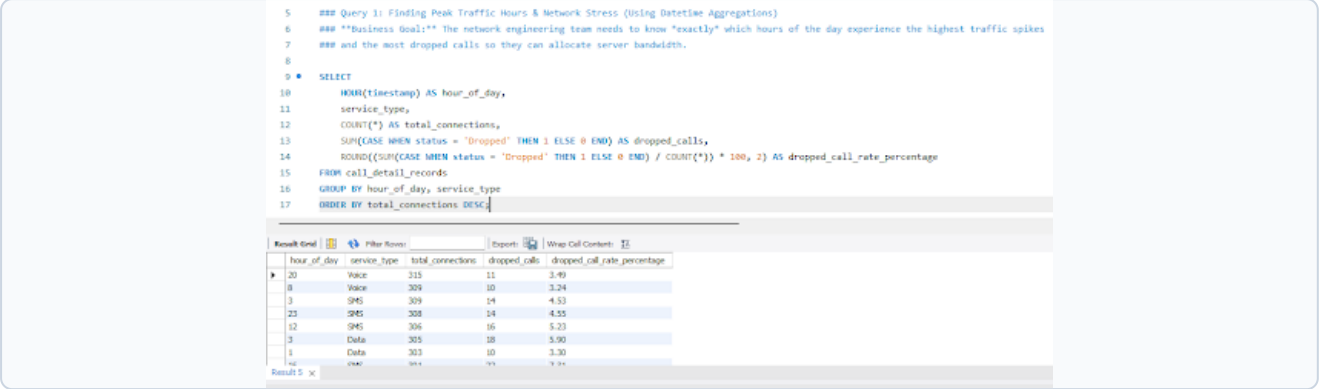
2. Deep-Dive SQL Analytics Engine

To support the executive dashboard, backend SQL workflows were designed to query network telemetry records, execute window functions, and join subscriber profile data.

Query 1: Peak Traffic Hours & Network Stress (Datetime Aggregations)

Aggregates network connection volume and dropped call counts by hour of the day and service type to help network engineers allocate server bandwidth effectively during peak hours.

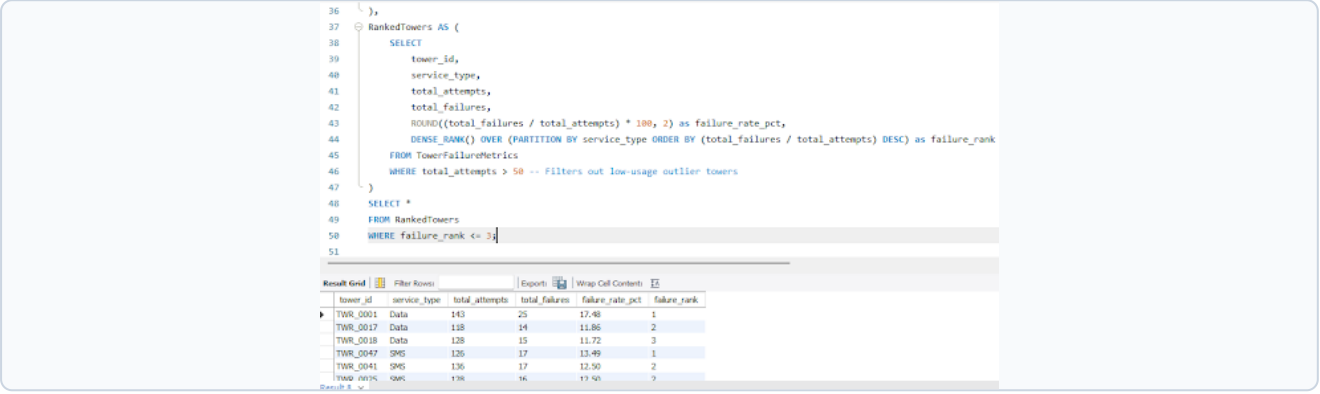
```
SELECT HOUR(timestamp) AS hour_of_day, service_type, COUNT(*) AS total_connections, SUM(CASE WHEN status = 'Dropped' THEN 1 ELSE 0 END) AS dropped_calls, ROUND((SUM(CASE WHEN status = 'Dropped' THEN 1 ELSE 0 END) / COUNT(*)) * 100, 2) AS dropped_call_rate_percentage FROM call_detail_records GROUP BY hour_of_day, service_type ORDER BY total_connections DESC;
```



Query 2: Tower Failure Rate Ranking (Window Functions)

Utilizes DENSE_RANK() partitioned across service categories to isolate the top 3 worst-performing cell towers based on failure percentage, filtering out low-usage outliers.

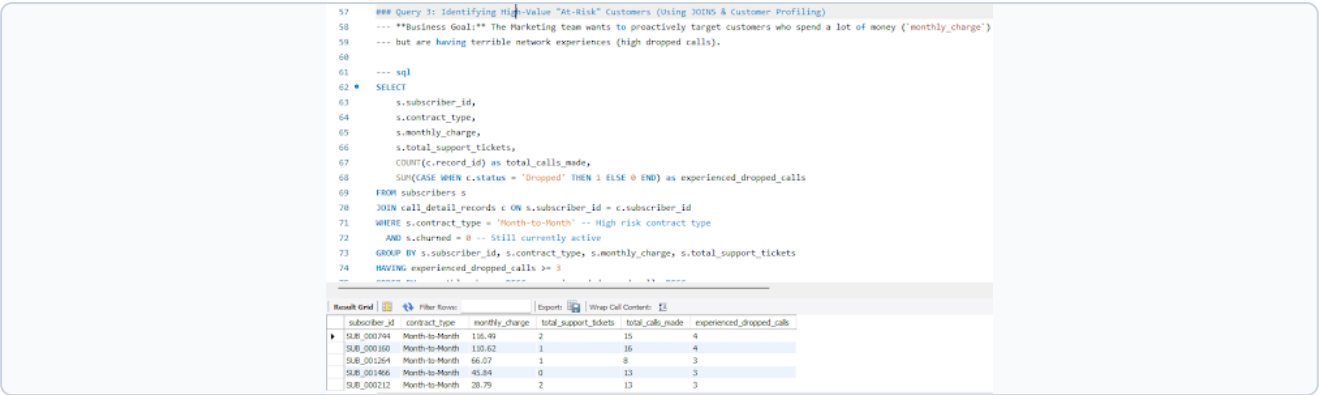
```
WITH RankedTowers AS ( SELECT tower_id, service_type, total_attempts, total_failures, ROUND((total_failures / total_attempts) * 100, 2) AS failure_rate_pct, DENSE_RANK() OVER (PARTITION BY service_type ORDER BY (total_failures / total_attempts) DESC) AS failure_rank FROM TowerFailureMetrics WHERE total_attempts > 50 -- Filters out low-usage outlier towers ) SELECT * FROM RankedTowers WHERE failure_rank <= 3;
```



Query 3: High-Value "At-Risk" Customer Identification (JOINS & Profiling)

Joins subscriber profile records with network call logs to pinpoint un-churned Month-to-Month customers who pay high monthly charges but suffer from 3 or more dropped calls.

```
SELECT s.subscriber_id, s.contract_type, s.monthly_charge, s.total_support_tickets,
COUNT(c.record_id) AS total_calls_made, SUM(CASE WHEN c.status = 'Dropped' THEN 1 ELSE 0 END) AS
experienced_dropped_calls FROM subscribers s JOIN call_detail_records c ON s.subscriber_id =
c.subscriber_id WHERE s.contract_type = 'Month-to-Month' AND s.churned = 0 GROUP BY
s.subscriber_id, s.contract_type, s.monthly_charge, s.total_support_tickets HAVING
experienced_dropped_calls >= 3 ORDER BY s.monthly_charge DESC;
```



3. Strategic Action Matrix & Operational Next Steps

Insight Focus	Key SQL / BI Finding	Operational Action Item
Peak Traffic Spikes	Voice & SMS traffic peaks around 8 PM (20:00) and 3 AM	Reallocate server bandwidth dynamically during peak congestion hours.
Tower Infrastructure	TWR_0001 & TWR_0017 reach up to 17.48% failure rate on Data	Dispatch emergency engineering teams for hardware inspection and re-tuning.
Subscriber Retention	High ARPU Month-to-Month users (e.g., SUB_000744 @ $116.49) face 4+ drops	Trigger proactive retention outreach, service credits, and priority support.